

Yash K. Jawale

POSTDOC · TU DELFT

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Research Interests

Cytoskeleton, Biophysics, Quantitative Biology, Synthetic Biology, Microscopy, Image analysis, Computation

Skills

Biochemical & Molecular Techniques	<i>in vitro</i> reconstitutions, encapsulation, protein purification, gene cloning
Experimental & Imaging Techniques	Microscopy (confocal, TIRF, IRM), microfluidics, optogenetics
Programming & Computational Tools	Python, MATLAB, Cytosim, ImageJ/Fiji, R, GROMACS
Other Technical Skills	LaTeX, FreeCAD (3D design), 3D printing, Blender (SciViz)

Publications

 Selection | Complete List   

- Roth, S., Gârlea, I. C., Vleugel, M., **Jawale, Y. K.**, Mulder, B. M., & Dogterom, M. (n.d.). *Reconstitution of basic mitotic spindles in cell-like confinement* [under review]
- Jawale, Y.K.** (2025, September). *Building Minimal Spindles: Reconstituting spindle positioning in synthetic cells* [Ph.D. dissertation]. Delft University of Technology. [10.4233/UUID:F4E25106-CB82-4706-90B7-88800D45C8D2](https://doi.org/10.4233/UUID:F4E25106-CB82-4706-90B7-88800D45C8D2)
- Van de Cauter, L.[‡], **Jawale, Y. K.**[‡], Tam, D., Baldauf, L., van Buren, L., Koenderink, G. H., Dogterom, M., & Ganzinger, K. A. (2024). High-speed imaging of giant unilamellar vesicle formation in cDICE. *ACS Omega*, 9(41), 42278–42288. [10.1021/acsomega.4c04825](https://doi.org/10.1021/acsomega.4c04825)
- Chaphalkar, A. R.[‡], **Jawale, Y. K.**[‡], Khatri, D., & Athale, C. A. (2021). Quantifying intracellular particle flows by DIC Object Tracking. *Biophysical Journal*, 120(3), 393–401. [10.1016/j.bpj.2020.12.013](https://doi.org/10.1016/j.bpj.2020.12.013)
- Joshi, Y. J., **Jawale, Y. K.**, & Athale, C. A. (2020). Modeling the tunability of the dual-feedback genetic oscillator. *Physical Review E*, 101(1). [10.1103/physreve.101.012417](https://doi.org/10.1103/physreve.101.012417)
- Jawale, Y. K.**, Rapol, U., & Athale, C. A. (2018). Open Source 3D printed focussing mechanism for cell phone based cellular microscopy. *Journal of Microscopy*, 273(2), 105–114. [10.1111/jmi.12765](https://doi.org/10.1111/jmi.12765)

[‡]Equal contributions

Research Experience

Doctoral work Asymmetric force generation in Synthetic cell Marileen Dogterom group	TU Delft, NL	Jan. 2020 - Present
Project Assistant Open Source Image Analysis of Sub-Cellular Microscopy Data: Leveraging Big Data Astronomy Tools	IISER Pune, IN	Aug. 2018 - Dec. 2019
Master's work Role of External Forcing in Directional Instability of Microtubule Transport Chaitanya Athale group (GPA 9.0 (of 10))	IISER Pune, IN	Jun. 2017 - Apr. 2018
Semester Project(s) <i>In vitro</i> Reconstitution and Quantification of recombinant Kinesin (KHC) Gliding Assay	IISER Pune, IN	Aug. 2016 - Apr. 2017
Semester Project(s) Optimizing the Foldscope: with PDMS and Glass lens microscopes, and Focusing Mechanism	IISER Pune, IN	Aug. 2015 - Apr. 2016
Summer Project iGEM (International Genetically Engineered Machine) Mycobacterium Revelio (Wiki) (Bronze Medal)	IISER Pune, IN	May 2015 - Sept. 2015

Education

PhD  Delft University of Technology (TUD)	Delft, NL	September 2025
BS-MS (Dual degree)  Indian Institute of Science Education and Research (IISER) Pune (CGPA 8.3 (of 10))	Pune, IN	May 2018

Conferences

 Selection (International)

EMBO Microtubules: From Atoms to Complex Systems (POSTER)
EMBO Frontiers in Cytoskeleton Research (POSTER)
SynCell International Conference on Engineering Synthetic Cells and Organelles
 (POSTER, TALK (2023))
ASCB | EMBO Cell Biology (POSTER)
NWO Dutch BioPhysics (POSTER)
DIVIDE 2 Spindle Structure: From Protein to Architecture (TALK & POSTER)
Naito 49 Frontiers of microtubule and its-related motors: atomic structures, cellular
 functions, development and diseases (TALK & POSTER)
HRZZ Mitotic spindle: From living and synthetic systems to theory (POSTER)
EMBO Creating is Understanding: Synthetic Biology Masters Complexity (FLASH TALK &
 POSTER)
iGEM Gaint Jamboree, International Genetically Engineered Machine (TALK & POSTER |
 BRONZE MEDAL)

EMBL Heidelberg, DE Jun. 2026 | 2024 | 2022 | 2020*
IISER Pune, IN Nov. 2025
Toulouse, FR | Minneapolis, USA | The Hague, NL
 Apr. 2024 | May 2023 | May 2022
Boston, USA Dec. 2023
Veldhoven, NL Oct. 2023 | 2022
CRG Barcelona, ES Sept. 2023
Sapporo, JP Jul. 2023
Dubrovnik, HR Apr. 2023
EMBL Heidelberg, DE Sept. 2019
Boston, USA Sept. 2015

Workshops

Lorentz Center Reconstituting Biology: Charting the Way to Minimal Cells
 Microscopy Image Analysis
Biophysical Society USA Modern Biophysical Tools and Techniques
 Dynamics of Complex Systems (Mathematical Biology)

Lieden, NL Jul. 2022
NCCS, Pune, IN Mar. 2019
IIT-Bombay, Mumbai, IN Dec. 2017
ICTS, Bangalore, IN May 2017

Teaching

Judge International Genetically Engineered Machine (iGEM)
 Assessing and grading the iGEM student teams at the Gaint Jamboree.
Teaching Assistant Physical Biology of the Cell II
 Designing assignments and conducting work-lectures.
Teaching Assistant Research Practical | Optical Tweezers
 Conducting practical and analysis sessions.
Teaching Assistant BioPhysics
 Designing assignments, exams and solutions.
Session Coordinator B4 Workshop on Bioimaging: Advanced Light Microscopy
 Conducted a MATLAB based image analysis session.
iGEM Team Mentor TB or not TB ([Wiki](#))
 Guiding the team with experimental design, modeling, analysis and presentation. Won Silver Medal.

Paris, FR Oct. 2022
TU Delft, NL 2021-2022
TU Delft, NL 2020-2021
TU Delft, NL 2020
IISER Pune, IN May 2019
IISER Pune, IN 2017

Achievements

INSPIRE Fellowship (MHRD, GOVT. OF INDIA)
GATE Life Sciences (AIR (ALL INDIA RANK): 188)
CSIR-UGC Life Sciences (LECTURESHIP (RANK: 35))

IN 2013-2018
IN 2018
IN 2017

Miscellaneous

Member Association for Socially Applicable Research ([ASAR](#))  ([Related works](#))
 Pro bono work - Data curation, analysis and visualizations.
European Synthetic Cell Initiative Highlight | Collaboration: Yash and Jacqueline ([blog](#))
iGEM Community Alumni Features | A Pursuit to Figuring Out How Biology Works: It Began at iGEM ([blog](#))
Invited Talk UGC HRDC Synthetic Biology: Refresher Course (Virtual)

IN Jun. 2018 - Present
NL Feb. 2023
IN Feb. 2023
IN Jan. 2021

Hobbies

• Numismatics • Philately • Homelabbing & self-hosting • Cycling